

11. (a) A solution contains one cation and up to three different anions. The anions possibly present are carbonate, chloride and sulfate.

(i) Devise a plan that unambiguously proves which anions are present in the mixture.

You should also give any observations and conclusions that enable you to identify the anions. [6 QER]

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- (ii) A student said that the cation in the mixture can only be sodium since all the possible anions form a soluble salt with sodium. Is he correct?

Justify your answer and state how you could prove if the statement were true. [2]

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- (b) A student is given four materials and asked to identify the **type of structure** present in each one by carrying out a series of tests.

She is told that the maximum temperature of a Bunsen burner flame is about 800 °C. She is also told that in at least one case, it will **not** be possible to come to a definite conclusion.

Her results are shown in the table below.

	A	B	C	D
Melting temperature / °C	100	>800	>800	>800
Solubility in water	soluble	insoluble	insoluble	soluble
Conductivity of solid	none	none	good	none
Conductivity of solution	none			good

- (i) Use the information in the table to identify each type of structure.
Where a definite conclusion cannot be reached explain your reasoning. [4]

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QUESTION CONTINUES ON PAGE 18



- (ii) For **one** of the materials where the type of structure could not be identified, suggest what further test(s) are needed to identify the type of structure. [2]

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- (iii) Suggest why it is difficult to identify a material as a metal when it is in powdered form. [1]

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END OF PAPER